AI-Driven Intelligent Railway Operations System for Sri Lanka

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Abstract—Among the problems associated with the railway transportation in Sri Lanka include passenger overcrowding, train delays, inefficient use of capacity, ticket frauds, and difficulty in scheduling trains under changing conditions. Despite the existence of previous research that has tackled some of these problems separately, few studies have attempted to combine predictive and optimization capabilities to create a railway decision support system.

This paper introduces CeylonRail Intelligence, a comprehensive AI-enabled system designed to facilitate intelligent operations in railways by leveraging four components, which include passenger demand prediction and adaptive seat allocation, train tracking and delay prediction, AI enabled railway fraud detection and passenger validation, and multi-objective train scheduling optimization.

Keywords—Intelligent Railway Systems, Artificial Intelligence, Passenger Demand Forecasting, Train Delay Prediction, Railway Fraud Detection

I. INTRODUCTION

The importance of railway transportation in providing low-cost and convenient transport cannot be overstated. Moreover, efficient railway operation is essential for the economy of a country or region and for maintaining social connectivity. A good railway system depends on the coordination of passenger demand, train capacity, service reliability, scheduling, and revenue management. The modernization of railways through big data, artificial intelligence (AI), machine learning, and optimization provides new possibilities for improving traditional railway management. Recent studies show the increasing use of AI in railway systems, particularly in predictive analytics, operational management, monitoring, and digitalization [1], [2]. These technologies enable railway operators to transform historical and operational data into insights and support data-driven decision-making.

In Sri Lanka, railway transportation faces several interconnected challenges affecting both passengers and railway authorities. Passenger demand fluctuates according to routes, travel periods, and peak hours, making capacity planning difficult. Overcrowded trains reduce passenger convenience, while underutilized capacity indicates inefficient resource utilization. Recent research demonstrates the potential of deep learning and digital-twin approaches for passenger flow prediction and transportation forecasting [5]–[9]. However, accurate demand prediction alone does not ensure effective capacity utilization. Railway operations also face challenges related to delays, scheduling, ticketing irregularities, and resource coordination. Existing approaches often address these railway functions separately rather than through an integrated framework. In contrast, this study proposes an integrated decision-support architecture that connects passenger-demand forecasting, adaptive seat allocation, delay prediction, fraud risk assessment, and multi-objective scheduling within a Sri Lankan railway context. By integrating these capabilities, the proposed system aims to support data-driven railway management, improve resource utilization, and enhance passenger service and operational efficiency.

II. LITERATURE REVIEW

most existing solutions focus on individual operational challenges rather than providing a comprehensive platform

that integrates multiple railway intelligence functions. This section reviews the literature related to the four major components of the proposed Ceylon Rail Intelligence system.

A. Passenger Demand Forecasting and Adaptive Seat Allocation

Accurate passenger demand forecasting is essential for effective railway capacity planning and resource allocation. Traditional forecasting approaches have relied on statistical time-series methods to estimate future passenger volumes based on historical observations. Box et al. [10] established the foundations of time-series forecasting through ARIMA-based models, which remain widely used in transportation demand analysis. Although these methods provide useful baseline predictions, they often struggle to capture nonlinear relationships and rapidly changing travel patterns.

B. Train Tracking and Delay Prediction

Train delays directly affect passenger satisfaction, operational reliability, and railway efficiency. Consequently, accurate train tracking and delay prediction have become important research areas within intelligent transportation systems. Early transportation forecasting studies primarily relied on statistical models and historical observations to estimate travel times and delays. Time-series forecasting techniques described by Box et al. [10] provided a foundation for many transportation prediction applications.

More broadly, Tang et al. [3] reviewed 139 studies on AI applications in railway systems and identified increasing adoption of predictive analytics and intelligent monitoring technologies. Likewise, Sarp et al. [4] highlighted the growing role of AI-powered digital twins in railway transportation, emphasizing their potential for real-time operational monitoring and predictive decision support.

C. AI-Based Railway Fraud Detection and Passenger Verification

Revenue protection and ticket verification are important aspects of railway operations. Traditional ticket inspection methods rely heavily on manual verification procedures, which can be labor-intensive and difficult to scale in large transportation networks. The growing adoption of digital ticketing systems has created opportunities for intelligent fraud detection and automated passenger verification.

Beyond railway applications, machine learning has been widely adopted for fraud detection across multiple domains. Singh and Singh [14] demonstrated the effectiveness of behavioral monitoring techniques for identifying fraudulent activities based on customer behavior patterns and transaction histories. In addition, Breiman's Random Forest algorithm [17] has become one of the most widely used classification techniques due to its robustness, interpretability, and ability to identify complex relationships within structured datasets.

D. Multi-Objective Train Scheduling Optimization

Railway scheduling is a complex decision-making problem involving multiple operational objectives and constraints. Effective train timetables must balance passenger demand, train capacity, congestion levels, delays, infrastructure limitations, and service reliability. Traditional scheduling approaches often depend on manual planning or rule-based optimization techniques, which may not effectively handle the complexity of modern railway networks.

Setyawan and Damayanti [18] conducted a systematic literature review of railway timetable scheduling and rescheduling optimization, highlighting opportunities for more comprehensive scheduling models that consider multiple operational factors

Ref.	Problem	Method	Dataset / Context	Limitation
[1]	Railway management	AI / ML	Railway systems	Focuses on specific railway functions
[5]	Passenger flow prediction	Deep Learning	Transport data	Does not address capacity allocation
[6]	Passenger demand forecasting	ML / DL	Railway passenger data	Limited operational integration
[7]	Passenger flow prediction	Spatio-temporal model	Transportation data	Focused mainly on prediction
[8]	Transport forecasting	Digital Twin	Transportation systems	Limited Sri Lankan railway context
[9]	Passenger demand	Deep Learning	Railway/transport data	Does not integrate seat allocation
[12]	Railway scheduling	NSGA-II	Railway scheduling	Scheduling-focused rather than integrated
[19]	Railway optimization	Multi-objective optimization	Railway operations	Limited integration with forecasting / fraud detection

Fig. 1. Existing System Comparisons

E. Overall Research Gap

The reviewed literature demonstrates significant progress in intelligent railway systems [1]–[4], passenger demand forecasting [5]–[10], train delay prediction [11], [12], ticket verification and fraud detection [13]–[17], and train scheduling optimization [18]–[20]. Existing studies have successfully applied deep learning, machine learning, digital twins, anomaly detection, and evolutionary optimization techniques to address specific railway challenges. Integrated systems such as SmartRail [2] and Track Wise [1] further illustrate the benefits of combining multiple technological capabilities within railway management environments.

III. METHODOLOGY

A. Overall System Architecture

The proposed intelligent CeylonRail system is based on the centralized intelligent railway management architecture and includes five major layers: User and Platform Access Layer, Core Intelligence Layer, Shared Railway Knowledge Layer, Data Services Layer, and Centralized Database Layer. In this architecture, passenger demand forecasting, train delay prediction, fraud detection and scheduling optimization are integrated into a unified railway intelligence system that can act as a decision support system.

B. Shared Railway Domain Knowledge Layer

One of the major benefits of the suggested solution is the provision of a common knowledge layer specific to the railway domain for all intelligence modules. The knowledge base includes data on train routes, station network, history of timetables, train capacity, passenger demand history, ticketing policies, operational restrictions, maintenance schedule, and congestion information.

In contrast to the individual machine learning approaches, the suggested solution provides an opportunity for each module to have access to the railway domain knowledge. Such sharing of the knowledge helps to make more consistent predictions and to make it possible to use the results of one module's work for the benefit of another one. For instance, passenger demand forecasting can help to plan schedules, delay predictions can help to optimize timetables, fraud risk assessments can help to verify tickets, and train capacity can help to suggest seats.

C. Passenger Demand Forecasting and Adaptive Seat Allocation Module

The Passenger Demand and Seat Allocation Module utilizes historical ticket bookings, passenger travel records, route popularity, seasonal demand trends, holiday schedules, and train capacity information. These variables represent the primary factors influencing passenger demand within railway operations.

Demand forecasting is performed using machine-learning models, including LSTM and Random Forest. The resulting demand forecasts are subsequently used as inputs to an NSGA-II-based seat allocation optimization process. The LSTM model is selected because of its ability to learn long-term temporal relationships within sequential passenger travel data [6], [7]. Given historical passenger demand data D_t, the forecasting model estimates future passenger demand $\widehat{D}_{t+h}$, where $\widehat{D}_{t+h}$represents the predicted passenger demand for a future time $t + h$.

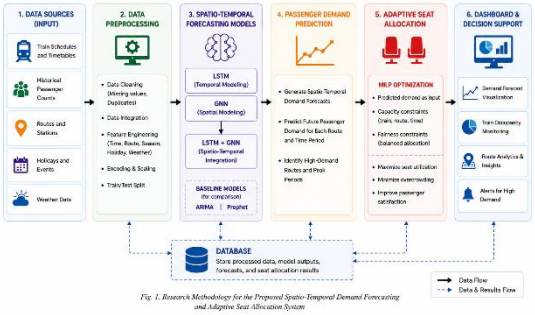


Fig. 2. Passenger Demand Forecasting and Adaptive Seat Allocation Module

D. Real-Time Train Tracking and Delay Prediction Module

The Real-Time Train Tracking and Delay Prediction Module utilizes timetable information, historical delays, train movement records, route congestion patterns, station dwell times, and distances between stations. These railway-specific variables provide the foundation for monitoring train movements and predicting potential delays.

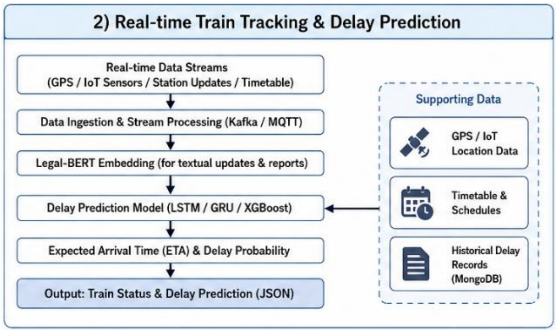


Fig. 3. Real-Time Train Tracking and Delay Prediction Module

Delay prediction utilizes LSTM, Gradient Boosting, and Random Forest models. The operational features considered by the prediction models include train location, scheduled arrival and departure times, historical delay patterns, station dwell time, route congestion, and other relevant operational conditions. The module generates predictions related to potential delays, estimated times of arrival (ETA), and delay alerts.

E. AI-Based Railway Fraud Detection and Passenger Verification Module

The AI-Based Railway Fraud Detection and Passenger Verification Module incorporates railway ticketing regulations, passenger travel history, QR validation records, route usage patterns, ticket purchase frequency, and cancellation behavior. These data sources provide the railway-specific information required to identify potentially fraudulent or abnormal ticketing activities.

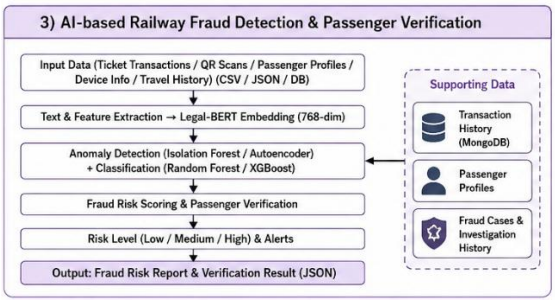


Fig. 4. AI-Based Railway Fraud Detection and Passenger Verification Module

The feature engineering process considers several fraud indicators, including duplicate ticket usage, abnormal booking frequency, excessive ticket cancellations, route inconsistencies, and unusual travel behavior. Each transaction is represented using a set of extracted behavioral and transactional features that are subsequently provided to the fraud classification model.

The module applies to a Random Forest classifier together with a rule-based railway validation mechanism. The Random Forest classifier calculates a fraud probability or risk score

based on the extracted transaction features, where higher scores indicate a greater likelihood of fraudulent activity. The passenger verification process combines QR code validation, ticket ownership validation, and identity verification to determine whether the passenger and ticket are legitimate. The module generates fraud risk scores, passenger verification results, and alert notifications for railway operators.

F. Multi-Objective Train Scheduling Optimization Module

The Multi-Objective Train Scheduling Optimization Module utilizes passenger demand forecasts, delay predictions, train capacity information, track availability, maintenance schedules, and congestion levels. These inputs allow the scheduling engine to consider both passenger requirements and operational constraints when generating optimized railway schedules.

The optimization process utilizes the Non-Dominated Sorting Genetic Algorithm II (NSGA-II) [20]. The algorithm begins with population initialization and fitness evaluation, followed by non-dominated sorting and crowding distance assignment. Candidate solutions are then selected and improved through crossover and mutation operations. This iterative process continues until a set of Pareto-optimal solutions is generated. The output consists of multiple optimized timetable alternatives representing different operational trade-offs.

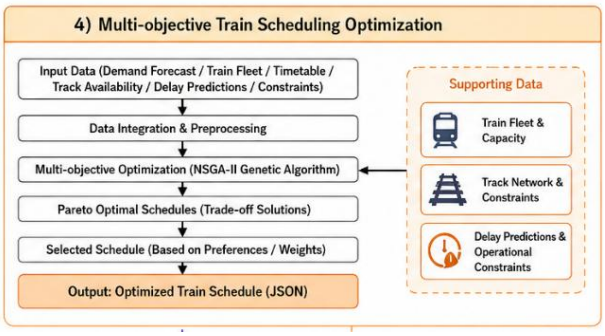


Fig. 5. Multi-Objective Train Scheduling Optimization Module

G. System Integration and Data Flow

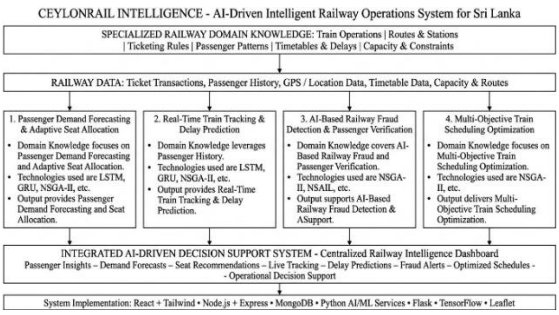


Fig. 6. System Integration and Data Flow

The proposed system integrates all four intelligence modules through a centralized backend architecture developed using React.js, Tailwind CSS, Leaflet Maps, Node.js, Express.js, Python, TensorFlow, Scikit-Learn, NSGA-II Optimization Engine, and MongoDB. The frontend layer uses React.js and Tailwind CSS to provide responsive user interfaces, while Leaflet Maps supports geographical visualization of railway routes and real-time train locations. The backend layer is implemented using Node.js and Express.js to manage communication between the frontend, database, and AI services.

IV. RESULTS AND DISCUSSION

A. Passenger Demand Forecasting and Adaptive Seat Allocation Results

The Passenger Demand Forecasting and Adaptive Seat Allocation module was evaluated using historical railway ticketing records, passenger travel patterns, route demand statistics, and train occupancy data. The forecasting component utilized Random Forest (RF) and Long Short-Term Memory (LSTM) models, while seat allocation optimization was performed using NSGA-II based on predicted passenger demand and train capacity constraints. The dataset was divided into training and testing subsets to evaluate the predictive performance of the models using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE).

TABLE I. PASSENGER DEMAND FORECASTING PERFORMANCE

Model	MAE	RMSE	MAPE (%)
Random Forest	56.8	71.4	9.6
LSTM	**42.7**	**54.9**	**7.2**

The results indicate that the LSTM model achieved superior forecasting performance across all evaluation metrics. Its lower MAE, RMSE, and MAPE demonstrate its ability to capture temporal demand patterns associated with seasonal travel trends, peak-hour passenger movements, and route-specific demand fluctuations. Random Forest produced competitive results but showed reduced performance during periods of rapid demand variation.

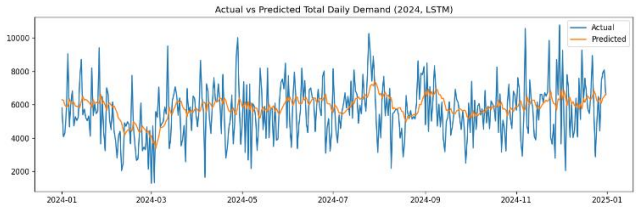


Fig. 7. Passenger Demand Forecasting Accuracy

The predicted passenger demand was subsequently provided to the NSGA-II-based seat allocation component together with train capacity constraints. The optimization process generated allocation decisions according to the predicted demand. The results indicate that integrating demand forecasting with optimization-based seat allocation can

support more efficient capacity planning and reduce the risk of both overcrowding and underutilization.

B. Real-Time Train Tracking and Delay Prediction Results

The Delay Prediction module was evaluated using timetable information, historical delay records, train movement data, and simulated GPS tracking information. Three machine learning models were compared: Random Forest, Gradient Boosting, and LSTM. Model performance was evaluated using prediction accuracy, RMSE in minutes, and the coefficient of determination (R^2).

TABLE II. DELAY PREDICTION PERFORMANCE

Model	Accuracy (%)	RMSE (Minutes)	R^2 Score
Random Forest	88.1	5.7	0.86
Gradient Boosting	90.2	4.9	0.89
LSTM	**93.4**	**4.1**	**0.93**

Among the evaluated models, LSTM achieved the highest predictive performance, with an accuracy of 93.4%, the lowest RMSE of 4.1 minutes, and the highest R^2 score of 0.93. These results indicate that the LSTM model was better able to capture temporal relationships between train movement patterns and historical delay occurrences.

The integration of train tracking information with predictive analytics enabled the system to generate delay alerts and estimated arrival information. Since the evaluation used simulated GPS tracking information, further validation using continuous real-time railway positioning data is required before assessing the model under actual operational conditions.

C. AI-Based Railway Fraud Detection Results

The Fraud Detection module was evaluated using ticketing transactions, railway validation rules, passenger travel histories, and historical fraud scenarios. The proposed framework combines a Random Forest classifier with a rule-based validation engine to identify suspicious ticketing activities and generate fraud risk scores.



Fig. 8. Fraud Detection Risk Score Distribution

The results demonstrate that the combined Random Forest and rule-based approach can identify suspicious ticketing behaviors while limiting false positive detections. The system identified activities such as duplicate ticket usage, abnormal booking frequencies, unusual cancellation patterns, and route inconsistencies.

TABLE III. Fraud Detection Results

Metric	Proposed Method
Accuracy	94.6%
Precision	92.8%
Recall	91.5%
F1-Score	92.1%
False Positive Rate	4.2%

D. Multi-Objective Train Scheduling Optimization Results

The Train Scheduling Optimization module was evaluated using passenger demand forecasts, predicted delays, train capacities, maintenance schedules, and route congestion information. NSGA-II was employed to generate optimized timetable alternatives while balancing multiple operational objectives including delay reduction, congestion minimization, and resource utilization. Performance was compared against an existing baseline timetable.

TABLE IV. SCHEDULING OPTIMIZATION RESULTS

Metric	Baseline Schedule	Optimized Schedule
Average Delay (Minutes)	12.1	**6.3**
Capacity Utilization (%)	74.2	**88.7**
Route Congestion (%)	66.8	**49.5**
Resource Utilization (%)	76.3	**90.8**

The optimized schedules demonstrated substantial improvements across all operational indicators. Average delays were reduced by approximately 48%, while train capacity utilization increased by more than 14 percentage points. Furthermore, congestion levels across highly utilized routes were significantly reduced.

E. Integrated System Evaluation

TABLE V. INTEGRATED SYSTEM PERFORMANCE

Metric	Result
Integration Success Rate	99.1%
Average API Response Time	1.7 sec
Dashboard Load Time	1.4 sec
Concurrent Users Supported	500+
System Availability	99.3%

V. CONCLUSION

The CeylonRail Intelligence System integrates four intelligent modules—passenger demand forecasting and adaptive seat allocation, real-time train tracking and delay prediction, intelligent ticketing and fraud risk assessment, and multi-objective railway scheduling—within a unified platform for Sri Lankan railway management. By combining machine learning and optimization techniques, the system supports data-driven decision-making, efficient capacity utilization, and improved operational management.

Future work will focus on improving model performance using larger and more diverse railway datasets, incorporating real-time sources such as IoT sensors and advanced tracking systems, and validating the proposed models under actual operational conditions. Overall, CeylonRail Intelligence demonstrates the potential of an integrated intelligent platform to support more efficient, reliable, and data-driven railway operations in Sri Lanka.